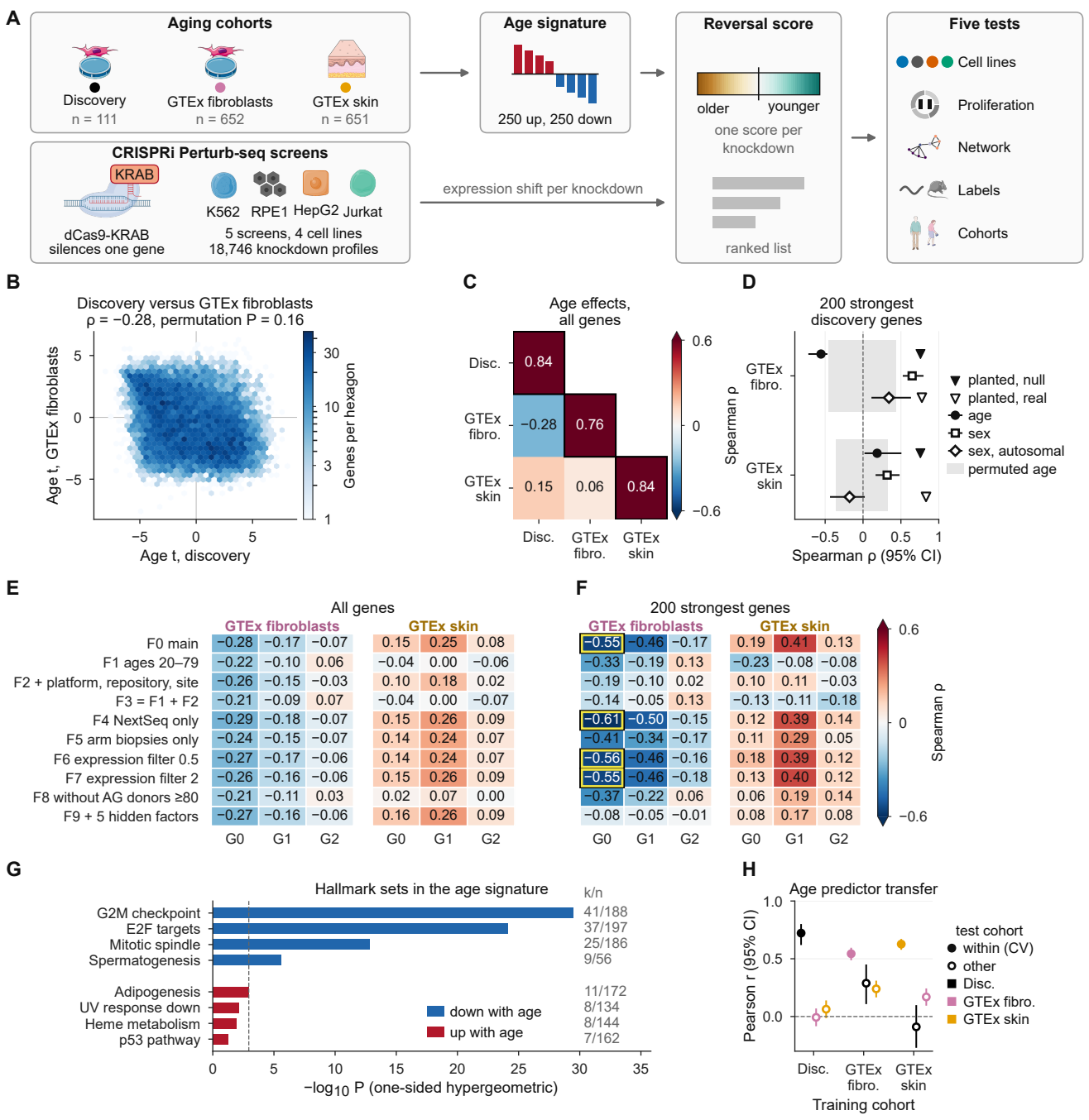

A
Aging cohorts
Discovery
n = 111
GTEx fibroblasts
n = 652
GTEx skin
n = 651
CRISPRi Perturb-seq screens
KRAB
dCas9-KRAB
silences one gene
K562
RPE1
HepG2
Jurkat
5 screens, 4 cell lines
18,746 knockdown profiles
Age signature
250 up, 250 down
expression shift per knockdown
Reversal score
older
younger
one score per
knockdown
ranked list
Five tests
Cell lines
Proliferation
Network
Labels
Cohorts
B
Discovery versus GTEx fibroblasts
ρ = −0.28, permutation P = 0.16
Age t, GTEx fibroblasts
Age t, discovery
Genes per hexagon
C
Age effects,
all genes
Disc.
GTEx fibro.
GTEx skin
0.84
−0.28
0.76
0.15
0.06
0.84
D
200 strongest
discovery genes
Spearman ρ
Spearman ρ (95% CI)
planted, null
planted, real
age
sex
sex, autosomal
permuted age
E
All genes
GTEx fibroblasts
GTEx skin
F0 main
F1 ages 20–79
F2 + platform, repository, site
F3 = F1 + F2
F4 NextSeq only
F5 arm biopsies only
F6 expression filter 0.5
F7 expression filter 2
F8 without AG donors ≥80
F9 + 5 hidden factors
G0
G1
G2
F
200 strongest genes
Spearman ρ
G
Hallmark sets in the age signature
k/n
G2M checkpoint 41/188
E2F targets 37/197
Mitotic spindle 25/186
Spermatogenesis 9/56
Adipogenesis 11/172
UV response down 8/134
Heme metabolism 8/144
p53 pathway 7/162
down with age
up with age
−log10 P (one-sided hypergeometric)
H
Age predictor transfer
Pearson r (95% CI)
Training cohort
test cohort
within (CV)
other
Disc.
GTEx fibro.
GTEx skin